# MEAL PLAN SELECTOR

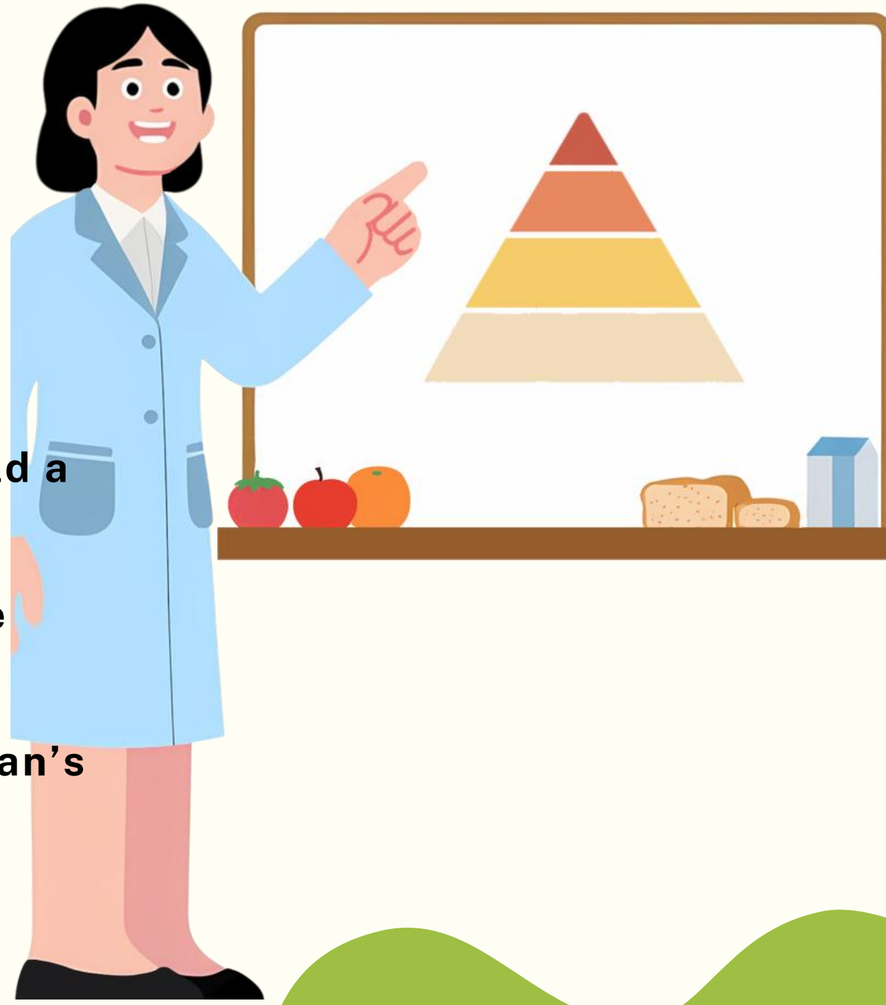
A Java console application using arrays, switch-case and loops

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Course: Java Programming



# ABSTRACT



- This project is a menu-driven Java program that lets a user build a daily vegetarian meal plan.
- Seven ready-made plans (breakfast, lunch and dinner each) are shown on screen.
- The user picks plans by number, and the program adds each plan’s cost to a running total.
- Meal names and prices are stored in arrays, the menu is printed with loops, and a switch-case handles each choice. Input is validated so invalid entries do not crash the program.

# INTRODUCTION

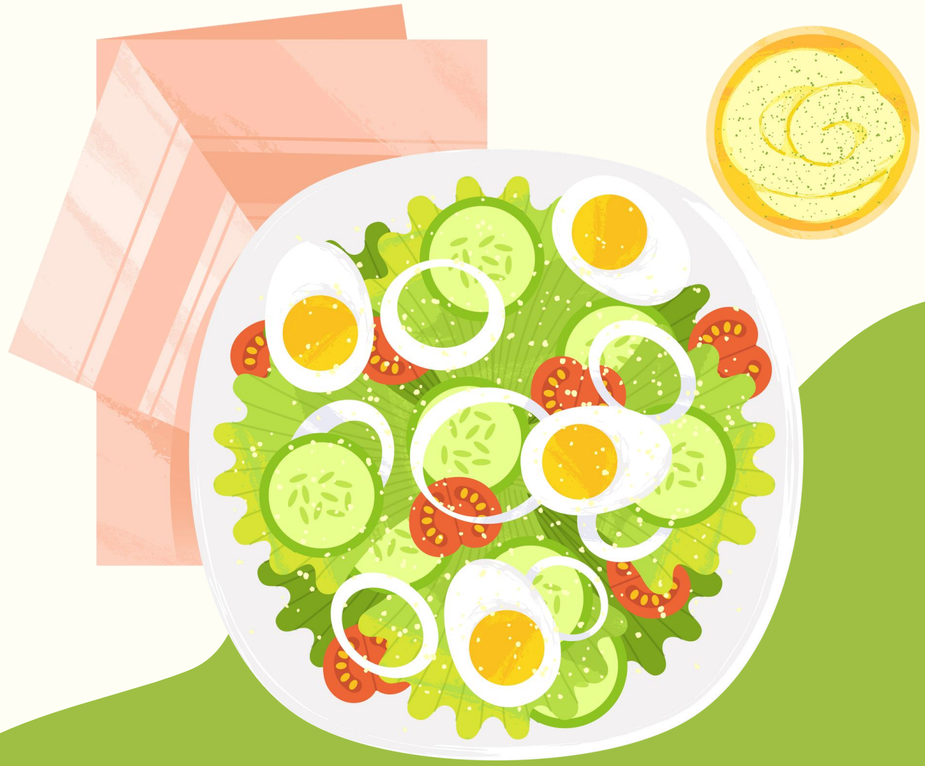
- Shows 7 meal plans plus an Eat out / NA option
- Tracks the total price as plans are added
- Repeats until the user exits

## Java topics used

- Scanner for user input
- 1D and 2D arrays
- do-while and for loops
- switch-case and if-else
- printf and ternary operator
- Input validation (hasNextInt)

# EXISTING SYSTEM

- Meals and prices are hard-coded, so any change means editing the code
- Cases 1 to 7 in the switch repeat the same logic
- All choices must be typed at once on one line, with no undo or edit
- Each plan has a single fixed price, with no quantities or budget limit
- Data is not saved, so the meal list and total are lost when the program exits



# SWITCH + ARRAYS

```
case 1:
    System.out.println("Selected Meal: 1 (" + meals[0][0] + ", ...)");
    totalPrice += prices[0];
    System.out.printf("Added Meal 1 to your list the price is %d%n", totalPrice);
    break;

case 2:
    System.out.println("Selected Meal: 2 (" + meals[1][0] + ", ...)");
    totalPrice += prices[1];
    System.out.printf("Added Meal 2 to your list the price is %d%n", totalPrice);
    break;

case 3:
    System.out.println("Selected Meal: 3 (" + meals[2][0] + ", ...)");
    totalPrice += prices[2];
    System.out.printf("Added Meal 3 to your list the price is %d%n", totalPrice);
    break;

case 4:
    System.out.println("Selected Meal: 4 (" + meals[3][0] + ", ...)");
    totalPrice += prices[3];
    System.out.printf("Added Meal 4 to your list the price is %d%n", totalPrice);
    break;
```

```
System.out.println("Meal Plan");
for (int i = 0; i < meals.length; i++) {
    System.out.print((i + 1) + ". ");
    for (int j = 0; j < labels.length; j++) {

        System.out.print(labels[j] + ": " + meals[i][j]);
        //?- is a terenery oprator which is a shorthand for an if-else statement.

        System.out.print(j < labels.length - 1 ? "; " : "\n");

        //  j = 0 (Breakfast) ,j = 1 (Lunch) ,j = 2 (Dinner)
       // so as long as j == 2 it will ; (continue) else it will end / go to new line (\n)
    }
}
```

# PROPOSED SYSTEM



- Arrays store labels, meals and prices in one place
- Nested for loops print the whole menu
- switch-case still handles choices 1 to 8
- Prices are read from the prices array
- hasNextInt() rejects non-numeric input
- do-while repeats until choice 8

```
int[] prices = {250, 280, 300, ...};

String[][] meals = {{b, l, d}, ...};

case 2:
totalPrice += prices[1];
break;
```

| CO | Outcome | Where i appears |
| --- | --- | --- |
| CO1 | Apply Java syntax, data types and operators | int, String, t operator, to arithmetic |
| CO2 | Use control structures to solve problems | do-while, fo switch-case |
| | | 1D arrays (l |

# RESULT SCREENSHOT

```
Enter all your choices on one line, separated by spaces (end with 8).
Example: 1 3 5 8
Your choices: 1 2 3 5 6 7 8

Selected Meal: 1 (paneer stuffed parathas with milk curd, ...)
Added Meal 1 to your list the price is 250
Selected Meal: 2 (large vegetable oats uttapams with coconut chutney and sambar, ...)
Added Meal 2 to your list the price is 530
Selected Meal: 3 (bowl of vegetable upma loaded with roasted cashews and peanuts, ...)
Added Meal 3 to your list the price is 830
Selected Meal: 5 (bowl of savory vegetable daliya porridge cooked in ghee with nuts, ...)
Added Meal 5 to your list the price is 1080
Selected Meal: 6 (aloo parathas served with a large bowl of Greek yogurt, ...)
Added Meal 6 to your list the price is 1330
Selected Meal: 7 (bowl of high protein sprouts chaat mixed with roasted peanuts, ...)
Added Meal 7 to your list the price is 1580
Eat out / NA selected. Ending your list.

Your budget is 1580
Thank you! Exiting...
```

Activate Windows
Go to Settings to activate Windows.

# THANK YOU

